CURVE-BASED INTEREST RATE RISK MEASURES

Learning Outcome

- explain why effective duration and effective convexity are the most appropriate measures of interest rate risk for bonds with embedded options

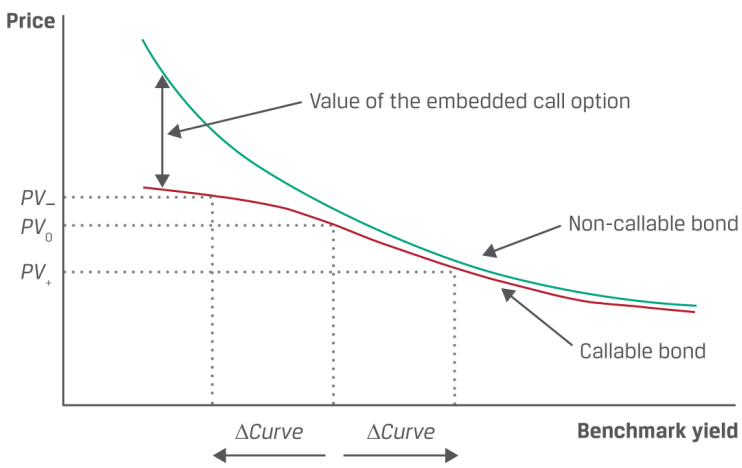
Yield duration and convexity assume a bond's cash flows are certain. However, if a bond has contingency features, such as embedded options, as with a callable (or puttable) bond, then future cash flows are *uncertain* since option exercise depends on the level of market interest rates relative to coupon interest being paid (or received). For example, the duration of a callable bond does *not* reflect the sensitivity of the bond price to a change in the yield-to-worst, since this represents only one of several possible outcomes based on future interest rates.

Consequently, bonds with embedded options do not have well-defined yields-to-maturity, so Macaulay and modified durations are not appropriate interest rate risk measures for such bonds. Rather, the appropriate measure of interest rate risk is the sensitivity of the bond's price to a change in a *benchmark yield curve*—for example, the government par curve—known as **effective duration**, a *curve* duration rather than a *yield* duration statistic.

Exhibit 1 shows the impact of an instantaneous change in the benchmark yield curve ($\Delta Curve$) on the price of a callable bond compared with that on a comparable non-callable bond. The two bonds have the same features (i.e., coupon rate, payment frequency, time-to-maturity, and credit risk). Note that the horizontal axis is a benchmark yield—a point on the par curve for government bonds.

Exhibit 1:

Interest Rate Risk Characteristics of a Callable Bond



▶ Expand / collapse Extended Description

The price of the non-callable bond is always greater than that of the comparable callable bond. The difference is the value of the embedded call option, which is held by the bond issuer (not the investor). When interest rates are high relative to the coupon rate, the value of the call option is low. When rates are low, the value of the call option is high since the issuer is more likely to exercise the option to refinance the debt at the lower prevailing rates. The investor bears the "call risk" since if the bond is called, the investor must reinvest the proceeds at a lower interest rate.

bonds are very similar. But when interest rates are low (Δ*Curve* is negative), the effective duration of the callable bond is lower than that of the otherwise comparable non-callable bond, since the callable bond price does not increase as much when benchmark yields fall; the presence of the call option limits price appreciation. Thus, an embedded call option reduces the effective duration of the bond, especially when interest rates are falling and the bond is more likely to be called. The lower effective duration can be interpreted as a shorter expected life—a reduced weighted average of time to receipt of cash flow.

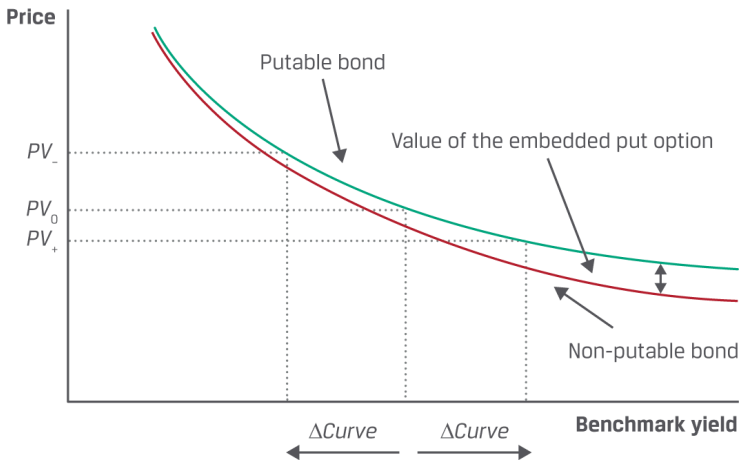
Analogous to effective duration, the second-order effect of a parallel shift in the benchmark yield curve is measured by **effective convexity**, a *curve convexity* statistic. In Exhibit 1, as the benchmark yield declines, the slope of the line tangent to the curve for the non-callable bond steepens, indicating positive convexity. But the slope of the line tangent to the callable bond flattens as the benchmark yield declines and reaches an inflection point, after which the effective convexity becomes negative.

When the benchmark yield is high and the value of the embedded call option is low, the callable and the non-callable bonds experience very similar effects from interest rate changes and both have positive convexity. But as the benchmark yield declines, the curves diverge, and at some point, the callable bond moves into the range of negative convexity, since the embedded call option has more value to the issuer and is more likely to be exercised. This situation further limits the potential price appreciation of the bond arising from lower benchmark yields.

Exhibit 2 shows characteristics of a bond with an embedded put option.

Exhibit 2:

Interest Rate Risk Characteristics of a Putable Bond



A putable bond allows the investor to sell the bond back to the issuer before maturity, usually at par value, protecting the investor from higher benchmark yields that would otherwise drive the bond's price below par. So, the price of a putable bond is always higher than that of an otherwise comparable non-putable bond, the price difference being the value of the embedded put option.

An embedded put option reduces the effective duration of the bond, especially when rates are rising. Given a parallel shift in the benchmark yield curve, if interest rates are low compared to the coupon rate, the value of the put option is low and the impact of a change in the benchmark yield on the bond's price is very similar to that for a non-putable bond. But when benchmark interest rates rise, the put option becomes more valuable to the investor, since the ability to sell the bond back to the issuer at par limits the price depreciation as rates rise. Importantly, putable bonds always have positive effective convexity.

covered in later lessons. MBS cash flows are contingent on homeowners' ability to refinance or pay off an existing mortgage with the proceeds from a new mortgage. Refinancing is a common feature of mortgages in the United States and some other jurisdictions that borrowers tend to exercise if interest rates fall.

Calculating effective duration (*EffDur*) is very similar to calculating approximate modified duration, as shown in Equation 1.

$$EffDur = \frac{(PV_-) - (PV_+)}{2 \times (\Delta Curve) \times (PV_0)} \qquad 1$$

The formula for calculating effective convexity (*EffCon*) is also very similar to the formula for approximate convexity, as shown in Equation 2.

$$EffCon = \frac{[(PV_-) + (PV_+)] - [2 \times (PV_0)]}{(\Delta Curve)^2 \times (PV_0)} \qquad 2$$

The differences are that Δ*Curve* is in the denominator—since effective duration is a *curve duration* statistic, it measures interest rate risk in terms of a parallel shift in the benchmark yield curve—and that PV_- and PV_+ are calculated using option pricing models. These models are covered in more detail in later modules but include such inputs as (1) the length of the call protection period, (2) the schedule of call prices and call dates, (3) an assumption about credit spreads over benchmark yields (which also includes any liquidity spread), (4) an assumption about future interest rate volatility, and (5) the level of market interest rates (e.g., the government par curve). The analyst holds the first four inputs constant and then raises and lowers the fifth input (i.e., parallel shifts) to derive PV_+ and PV_-, respectively.

Although effective duration and effective convexity are the appropriate interest rate risk measure for bonds with embedded options, they may also be useful with option-free bonds to supplement yield duration. Importantly, small differences may arise between a bond's effective duration and its modified duration because when the government par curve is shifted in the model, the government spot (i.e., zero) curve is also shifted, but not in the same parallel manner. Thus, the change in the bond price is not the same as it would be if its yield-to-maturity changed by the same amount as the change in the par curve.

In general, modified duration and effective duration on an option-free bond are not identical. However, the difference narrows when the yield curve is flatter, the time-to-maturity is shorter, and the bond is priced closer to par value; the difference disappears only in the rare circumstance of a flat yield curve.

For example, consider BRWA's five-year, 3.2% (semiannual) coupon bond priced at par for settlement on 15 October 2025 and maturity on 15 October 2030. Using PV_0 = 100.00 and Δ*Yield* = 0.0005, in an earlier lesson, we found

PV_+ = 99.771 and PV_- = 100.230,

resulting in yield duration and convexity statistics as follows:

ApproxModDur = 4.587 and *ApproxCon* = 24.239.

Instead of a change in the bond's yield-to-maturity, assume there is an upward and downward shift of the benchmark government par curve (Δ*Curve*) by 5 bps, which causes spot rates to move in a slightly non-parallel manner. Using PV_0 = 100.00 and Δ*Curve* = 0.0005 and assuming the following new prices,

PV_+ = 99.760 and PV_- = 100.241,

then using Equation 1, effective duration is 4.816:

$$EffDur = \frac{(PV_-) - (PV_+)}{2 \times (\Delta Curve) \times (PV_0)}$$

$$EffDur = \frac{(100.241) - (99.760)}{2 \times (0.0005) \times (100)}$$

EffDur = 4.816

$$EffCon = \frac{[(PV_{-}) + (PV_{+})] - [2 \times (PV_0)]}{(\Delta Curve)^2 \times (PV_0)}$$

$$EffCon = \frac{[(100.241) + (99.760)] - [2 \times (100)]}{(0.0005)^2 \times (100)}$$

EffCon = 40.000

EXAMPLE 1

Effective Duration and Convexity

The portfolio manager asks you, the analyst on a fixed-income team, to determine the interest rate sensitivity of a callable bond she is considering. The portfolio manager is seeking to invest in a bond with a duration between 7.0 and 8.0 and positive convexity.

Suppose the full price of the callable bond is 101.060 per 100 of par value. When the government par curve is raised and lowered by 25 bps, the new full prices for this callable bond from your option valuation model are 99.050 and 102.891, respectively. Therefore,

- PV_0 = 101.060,
- PV_+ = 99.050,
- PV_- = 102.891, and
- $\Delta Curve$ = 0.0025.

You determine effective duration and effective convexity for this callable bond as follows:

$$EffDur = \frac{(PV_{-}) - (PV_{+})}{2 \times (\Delta Curve) \times (PV_0)}.$$

$$EffDur = \frac{(102.891) - (99.050)}{2 \times (0.0025) \times (101.060)}.$$

EffDur = 7.601.

$$EffCon = \frac{[(PV_{-}) + (PV_{+}) - 2 \times (PV_0)]}{(\Delta Curve)^2 \times (PV_0)}.$$

$$EffCon = \frac{[(102.891) + (99.050)] - [2 \times (101.060)]}{(0.0025)^2 \times (101.060)}.$$

EffCon = −283.

You recommend that the portfolio manager not buy this callable bond. While its effective duration meets her criteria, the bond has negative effective convexity.

Question Set

1. Match each scenario to the interest rate risk measures below.

Scenario	Interest rate risk measure
A. Bond's percentage price change given a change in its yield-to-maturity	Money duration
B. Linear approximation of the bond's price change in currency, given a 1% change in yield	Modified duration
C. Callable bond	Effective duration

Hide Solution

The match for A is modified duration. Modified duration is used to estimate the percentage price change given a change in yield-to-maturity.

The match for B is money duration. Money duration expresses modified duration in currency units, which can be used to estimate the currency unit change in the position of a bond.

The match for C is effective duration. Callable bonds have uncertain future cash flows with respect to future interest rates. Therefore, a curve duration statistic, such as effective duration, would be best to measure interest rate risk.

Questions 2–5 relate to the following information:

As a member of a fixed-income team, you are asked to examine pricing data for a Viviyu Inc. callable bond. The current full price, the price assuming the benchmark yield curve shifts up 25 bps, and the price assuming the benchmark yield curve shifts down by 25 bps, are as follows:

- PV_0 = 102.208,
- PV_+ = 100.004, and
- PV_- = 103.891.

2. Calculate the effective duration of the Viviyu Inc. callable bond.

Hide Solution

Solution:

$$EffDur = \frac{(PV_-)-(PV_+)}{2 \times (\Delta Curve) \times (PV_0)}.$$

$$EffDur = \frac{(103.891)-(100.004)}{2 \times (0.0025) \times (102.208)}.$$

EffDur = 7.6061.

3. In an audit of the data, it was determined that PV_+ was incorrect and that the correct value of PV_+ is 100.50426. All other data values were correct. Explain whether the updated data would increase or decrease the value of effective duration.

Hide Solution

Solution:

There is no need to recalculate the effective duration. Since the updated PV_+ value is larger than the prior one, with all else constant, the numerator would be lower; thus, effective duration would decrease.

4. Calculate the effective convexity of the Viviyu Inc. callable bond using the following:

- PV_0 = 102.208,
- PV_+ = 98.504,
- PV_- = 103.891, and
- $\Delta Curve$ = 0.0025.

Hide Solution

Solution:

$$EffCon = \frac{[(PV_-)+(PV_+)]-[2 \times (PV_0)]}{(\Delta Curve)^2 \times (PV_0)}.$$

$$EffCon = \frac{[(103.891)+(98.504)]-[2 \times (102.208)]}{(0.0025)^2 \times (102.208)}.$$

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5. In Question 4, the effective convexity was negative, not positive as it was for the effective duration. Explain what negative effective convexity implies.

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Solution:

Negative effective convexity implies that the bond price will decline by a greater amount with an upward shift in benchmark yield than it will increase with a decline in the benchmark yield.

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MD For example 1, I believe the change in the yield curve should be 0.0025, not 0.00025, as it is 25 basis points (0.25%). It seems it is only an error on value display, as the results for both Effective Duration and Effective Convexity do match using 0.0025.

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